

Session Objectives

1. Describe the risk factors for developing heart failure.
2. Identify types and stages of heart failure: A → D; HFrEF, HFpEF; right- and left-sided heart failure, acute and chronic heart failure.
3. Apply and interpret appropriate tests for heart failure diagnosis.
4. Review the clinical manifestations of heart failure.
5. Apply guideline-directed medical therapy (pharmacologic and non-pharmacologic) in management of heart failure across the spectrum of stages and classes.
6. Explain the importance of improving care coordination amongst interprofessional team members to decrease mortality of patients with heart failure.

Case

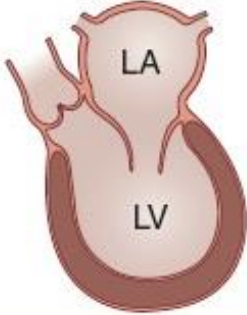
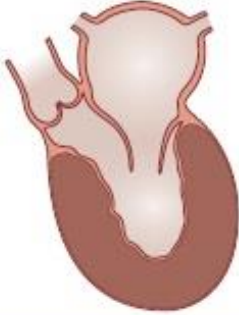
- A 65-year-old man with a history of HTN, hyperlipidemia, and tobacco use reports to the clinic because of 2 weeks of increasing dyspnea on exertion. He now sleeps on 4 pillows at night to aid in breathing. He has T2DM variably controlled with metformin and diet, and drinks 3-4 glasses of wine/day. Physical examination: jugular venous distention, an S_3 heart sound, and bilateral LE edema. ECG shows Q waves in leads V_2 - V_4 . He gives a history of hospitalization for "issues with my heart" about a year ago when he had sudden chest pain that radiated to his left arm, after which he had been asymptomatic.
- Which of the following is the most likely cause of this patient's condition?
 - A. Alcohol
 - B. Aortic stenosis
 - C. Coronary artery disease
 - D. Coronavirus
 - E. Coxsackievirus B
 - F. Sarcomere gene mutation



Heart Failure is a Clinical Syndrome

- Heart is unable to pump blood at a rate sufficient to meet metabolic demands leading to symptoms of fatigue, shortness of breath and volume overload
- Abnormalities in cardiac function and neurohormonal regulation, which are accompanied by effort intolerance, fluid retention, and a reduced longevity
- Complex clinical syndrome that can result from any structural or functional cardiac disorder that impairs the ability of ventricle to fill or pump

Types of Heart Failure: by Structure

	Dilated Cardiomyopathy	Hypertrophic Cardiomyopathy	Restrictive Cardiomyopathy
Ventricular morphology	Dilated LV with little concentric hypertrophy 	Marked hypertrophy, often asymmetric 	Fibrotic or infiltrated myocardium 
Etiology examples	Genetic, inflammatory, toxic, metabolic, peripartum	Genetic	Amyloidosis, genetic
Symptoms	Fatigue, weakness, dyspnea, orthopnea, PND	Dyspnea, angina, syncope	Dyspnea, fatigue
Physical exam	Pulmonary crackles, S3; if RV failure present: JVD, hepatomegaly, peripheral edema	S4; if outflow obstruction present: systolic murmur loudest at the left sternal border, accompanied by mitral regurgitation	Predominantly signs of RV failure: JVD, hepatomegaly, peripheral edema
Pathophysiology	Impaired systolic contraction	Impaired diastolic relaxation; LV systolic function vigorous, often with dynamic obstruction	"Stiff" LV with impaired diastolic relaxation but usually normal systolic function
Cardiac size on chest radiograph	Enlarged	Normal or enlarged	Usually normal
Echocardiogram	Dilated, poorly contractile LV	LV hypertrophy, often more pronounced at the septum; systolic anterior movement of MV with mitral regurgitation	Usually normal systolic contraction; the myocardium may have enhanced echogenicity in infiltrative disorders

LV, left ventricle; PND, paroxysmal nocturnal dyspnea; RV, right ventricle; JVD, jugular venous distention; MV, mitral valve.

Types of Heart Failure: Left vs. Right

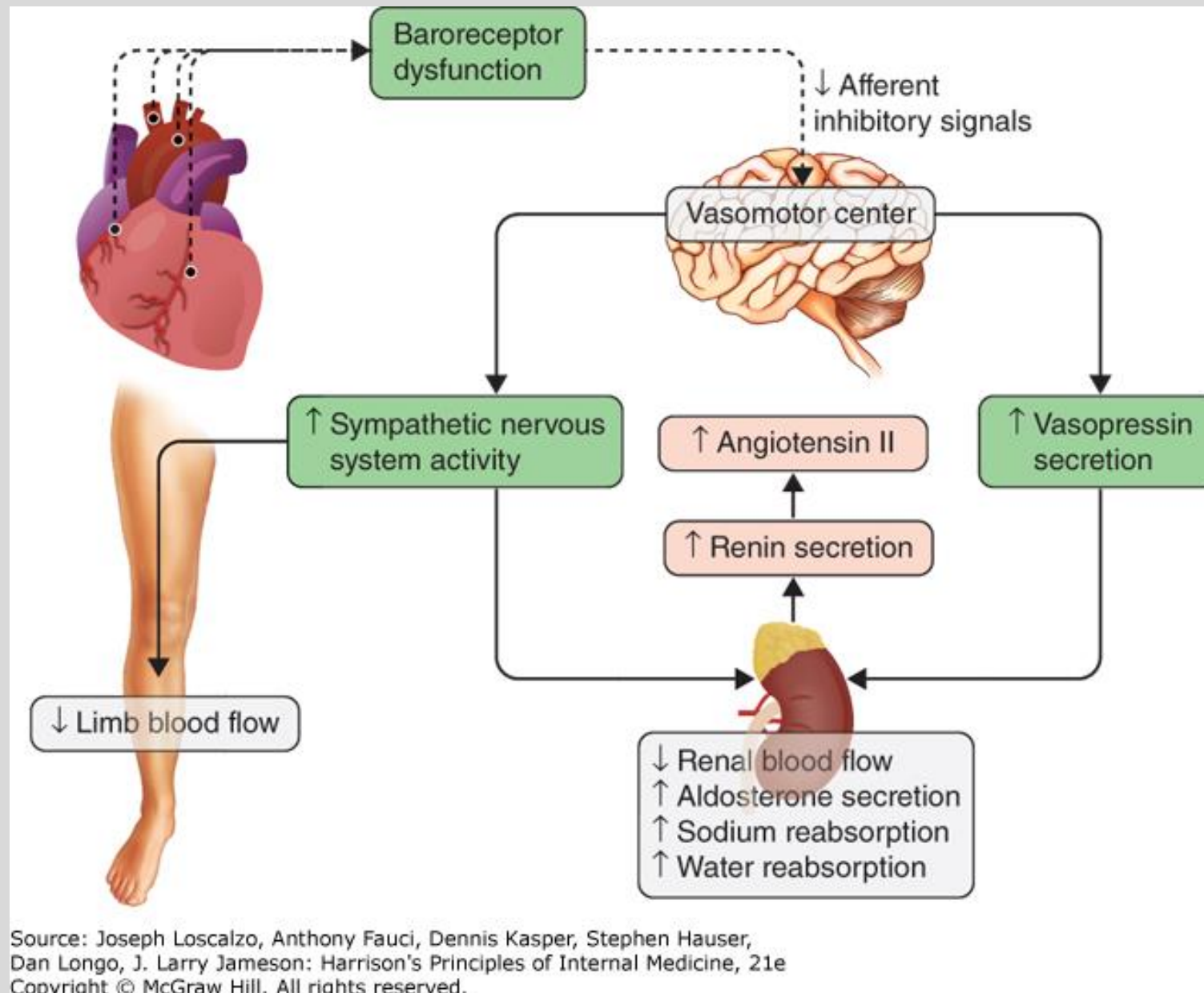
Left-side Heart Failure:

- pump failure:
 - HFrEF and HFpEF

Right-side heart failure:

- Most commonly caused by left-sided HF
 - Cor pulmonale: isolated right heart failure, infrequent, secondary to lung pathology

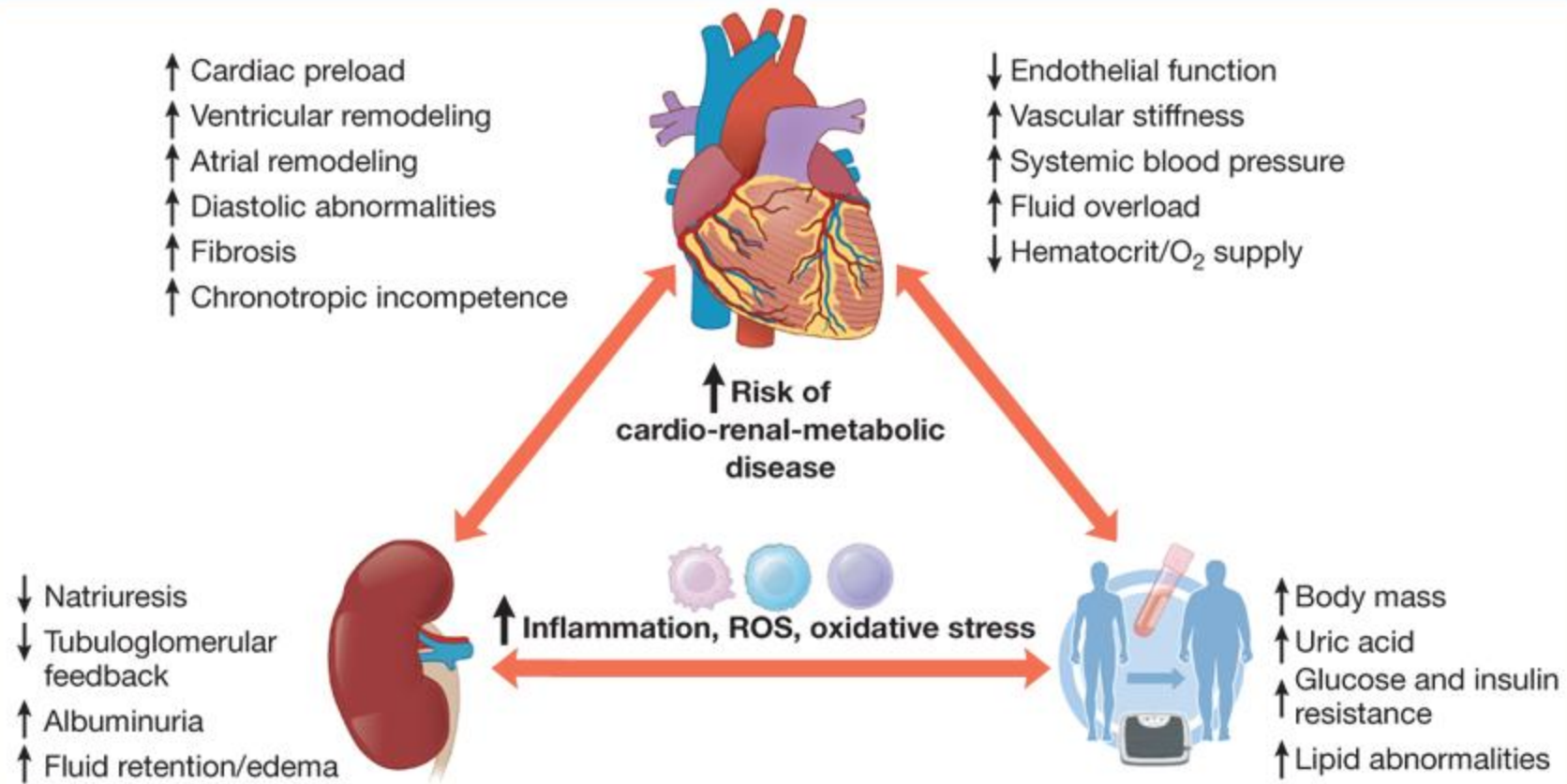
Pathophysiology: Neurohormonal Activation



- $BP = CO \times SVR$
 - Adrenergic Nervous System
 - Renin-Angiotensin-Aldosterone System
 - Increased production of Antidiuretic Hormone (ADH)

Pathophysiology of HFpEF

Heart failure with preserved ejection fraction: multiorgan systemic pathophysiology



- Impaired diastolic relaxation of the LV
- Vascular stiffness
- Renal dysfunction
- Metabolic inflammation
- Oxidative stress and impaired nitric oxide signaling

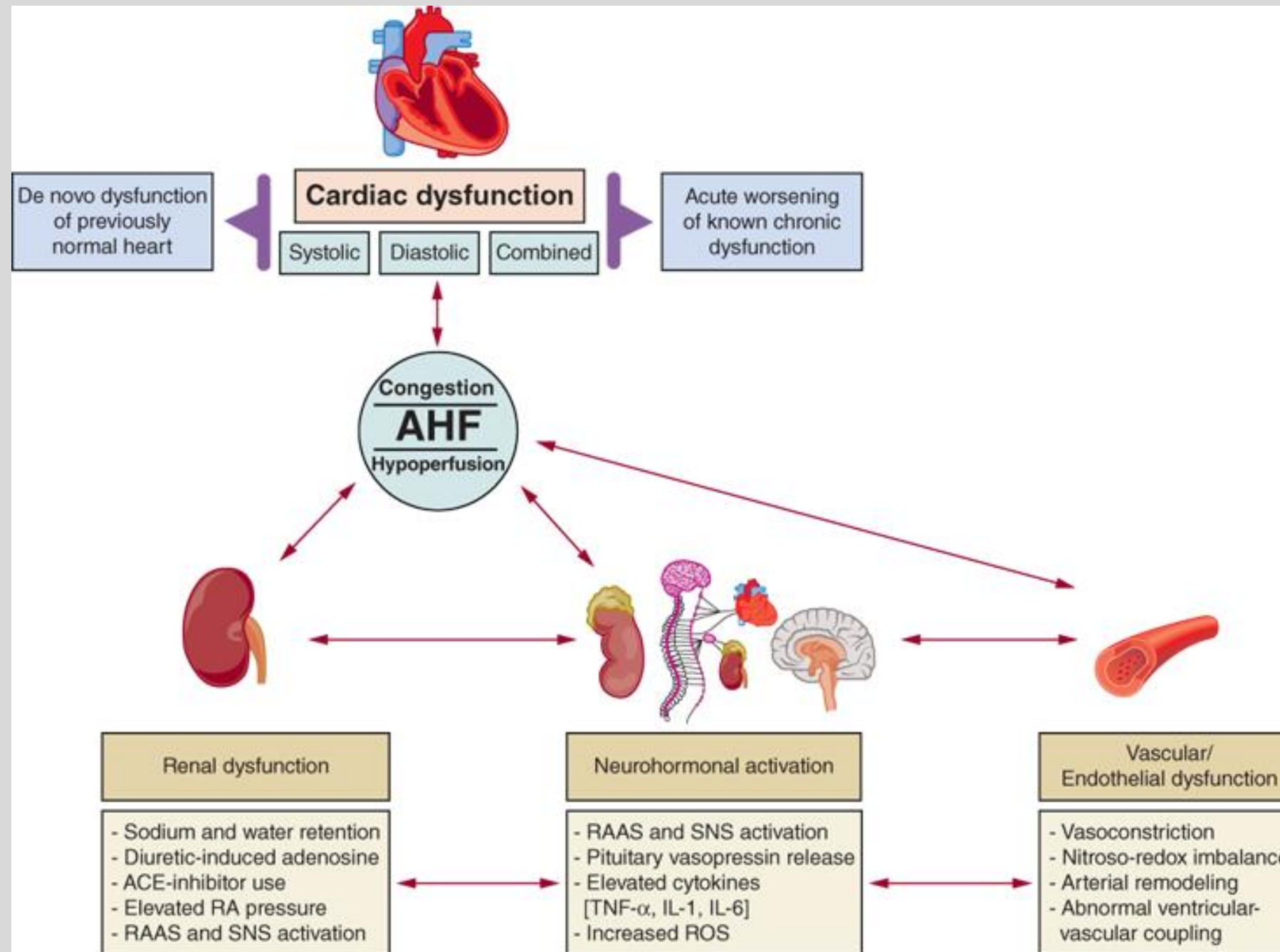
Chapter 49 Fuster and Hurst's Central Illustration. Patients with heart failure with preserved ejection fraction have increased risk of cardio-renal-metabolic disease. ROS, reactive oxygen species.

Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
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Ferreira J, Sharma K, Butler J. Diagnosis and Management of Heart Failure with Preserved Ejection Fraction. In: Fuster V, Narula J, Vaishnava P, Leon MB, Callans DJ, Rumsfeld J, Poppas A. eds. *Fuster and Hurst's The Heart*, 15e. McGraw Hill; 2022.

Givertz MM, Mehra MR. Heart Failure: Pathophysiology and Diagnosis. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. eds. *Harrison's Principles of Internal Medicine*, 21e. McGraw-Hill Education; 2022.

Pathophysiology: Acute HF



Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
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Right-Sided Heart Failure

Cardiac causes

Left-sided heart failure

Pulmonic valve stenosis

Right ventricular infarction

Pulmonary parenchymal diseases

Chronic obstructive pulmonary disease

Interstitial lung disease (eg, sarcoidosis)

Chronic lung infection or bronchiectasis

Pulmonary vascular diseases

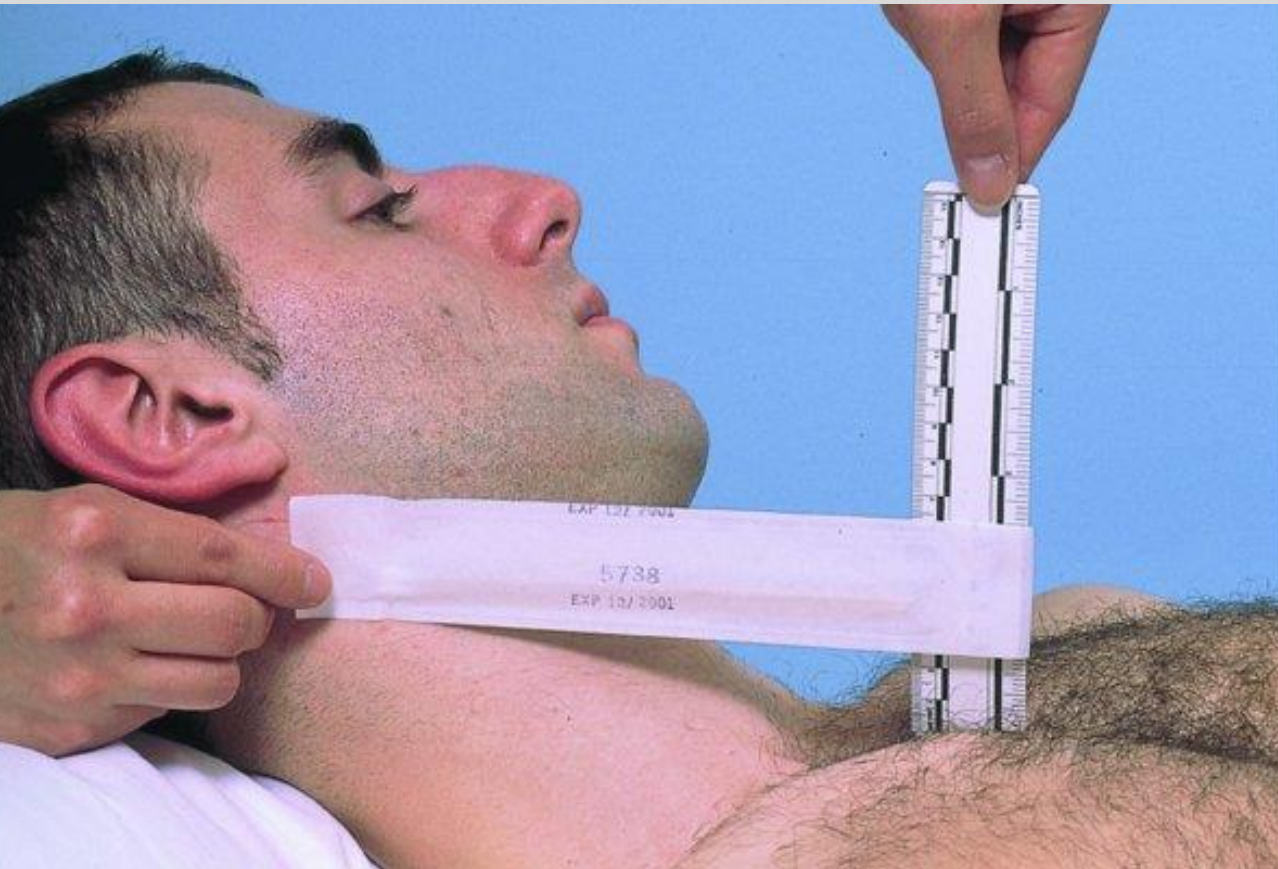
Pulmonary embolism

Pulmonary arteriolar hypertension

Physical Examination

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Measure the JVP with a horizontal
card and vertical ruler

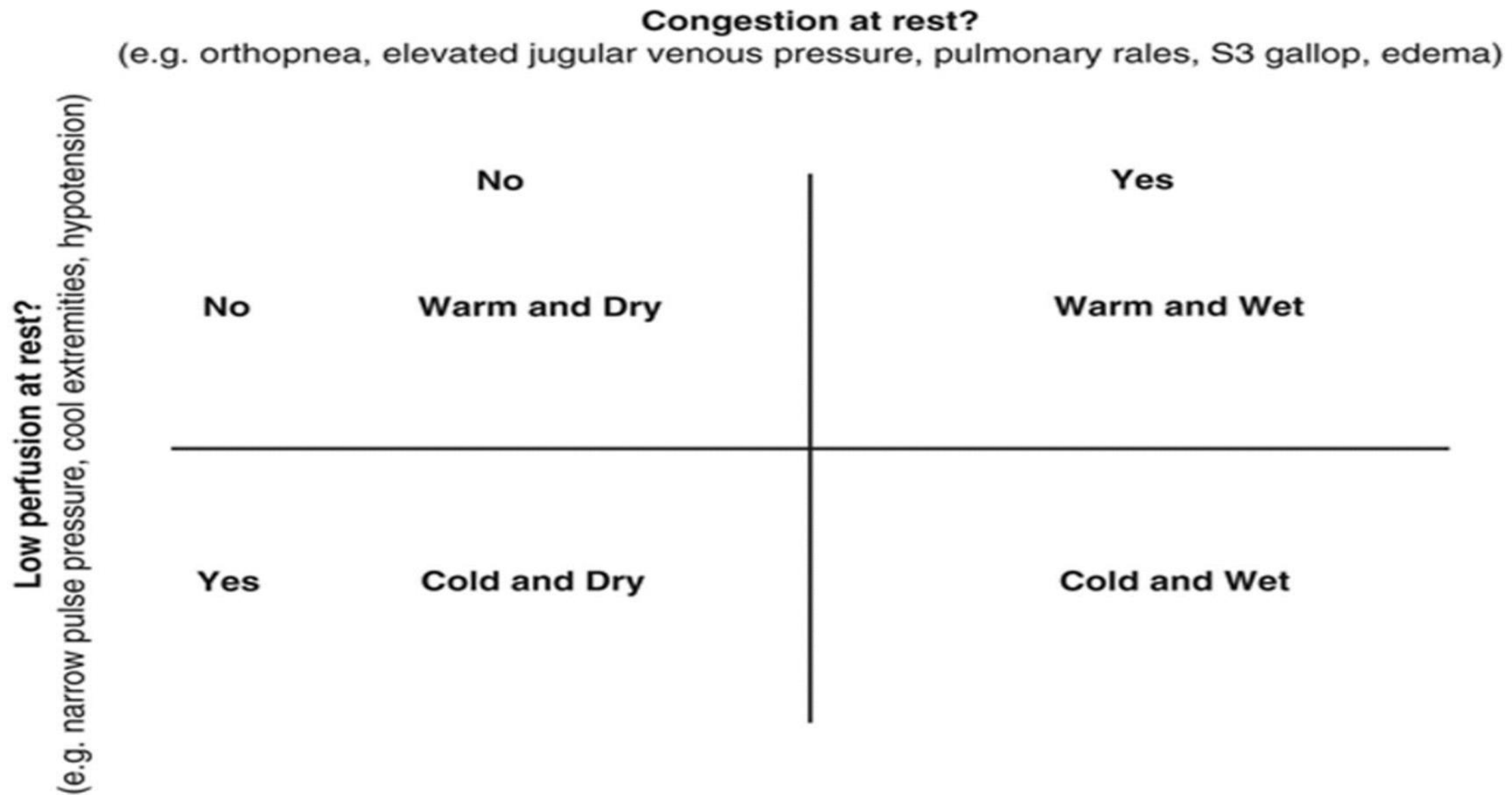


Pretibial
Edema

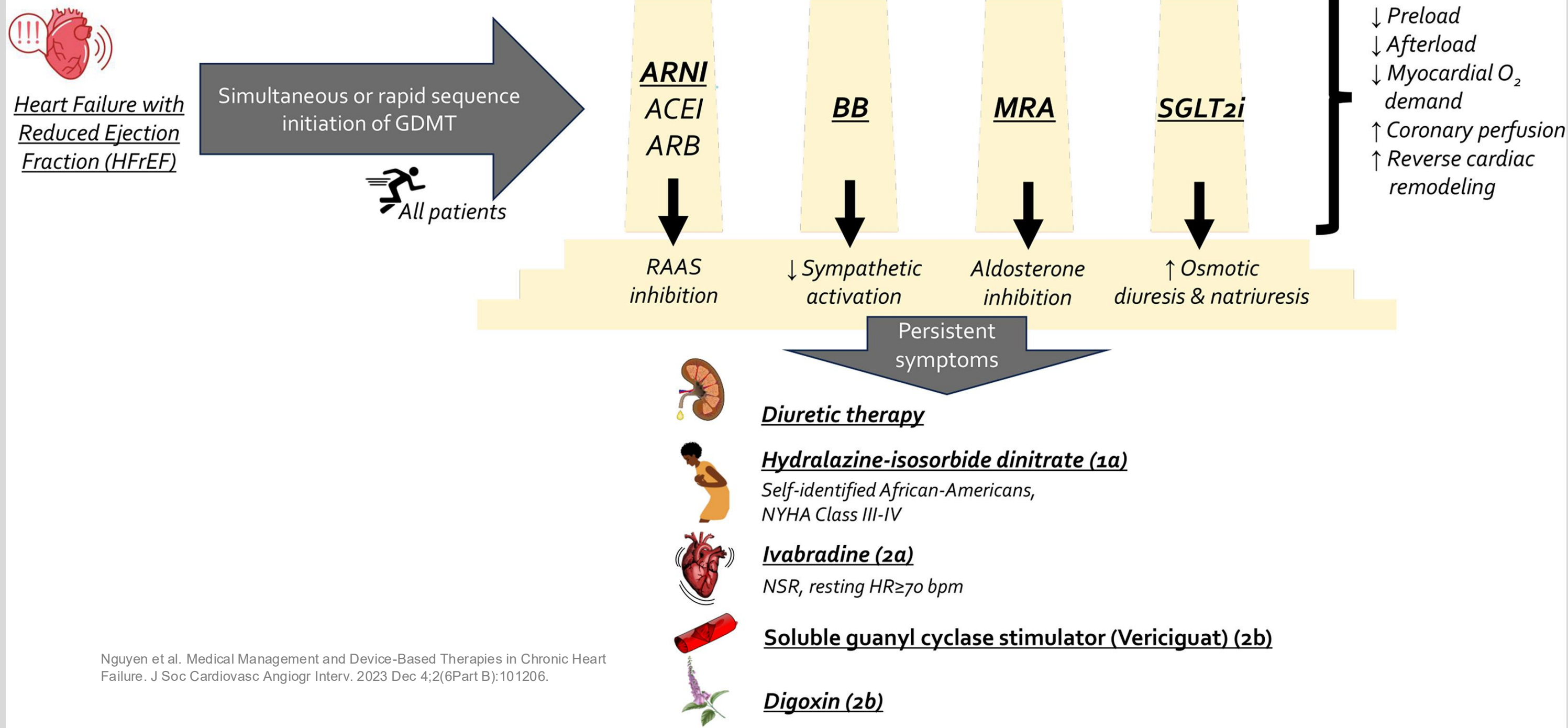
Diagnosis of Heart Failure

- Chest X ray
- ECG
- Labs (CBC, CMP, TFTs, HIV, lipids, A1C, COVID-19), natriuretic peptide
- Ischemic - Troponins, Coronary Angiogram
- Echo - dilated, valve disease, infiltrative cardiomyopathies
- Cardiac MRI
- Arrhythmia
- Family history
- Myocarditis
- Chronic HF with acute decompensation (all of the above plus):
 - Infection;
 - Medication or dietary non-adherence;
 - Toxins (NSAIDs, EtOH);
 - Arrhythmia

Acute Decompensated Heart Failure



HFrEF: Management



SODIUM-HF

Study Objective

Assess the effects of lowering the amount of sodium in the diet on clinical outcomes in a heart failure (HF) population.



Procedures



12 month intervention period
+ patients followed for additional
12 months



Dietary sodium intake assessed
using a 3-day food record.

Participants



806
Patients



26
Sites

Patients randomized to the low sodium diet followed a dietary intervention designed to reduce sodium intake to <1500 mg/day. Patients randomized to usual care followed local guidelines.

6 Countries



Results



In patients with HF, a dietary intervention to reduce dietary sodium intake did not reduce clinical events, such as being hospitalized or going to the ER.



Modest improvements were seen in quality of life and New York Heart Association HF class.



SODIUM-HF is the largest trial of its type and provides a major update to published evidence.

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SODIUM-HF



Ezekowitz JA, Colin-Ramirez E, Ross H, et al. Reduction of dietary sodium to less than 100 mmol in heart failure (Sodium-hf): an international, open-label, randomised, controlled trial. *Lancet*. 2022;399(10333):1391–1400.

GDMT: Left Ventricular Assist Device (LVAD)

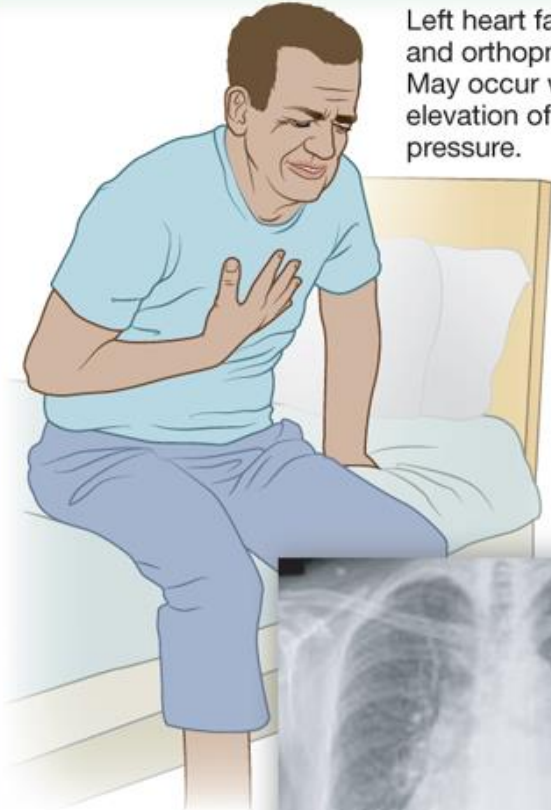
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Common transitions in advanced systolic heart failure

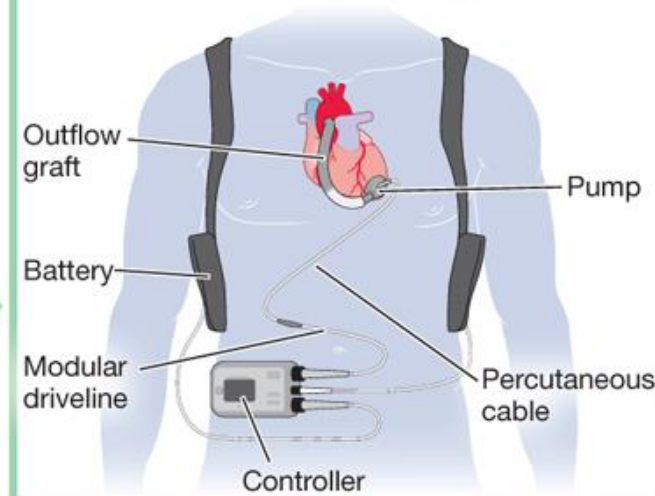
Worsening of heart failure

Left heart failure: dyspnea and orthopnea; May occur with or without elevation of central venous pressure.

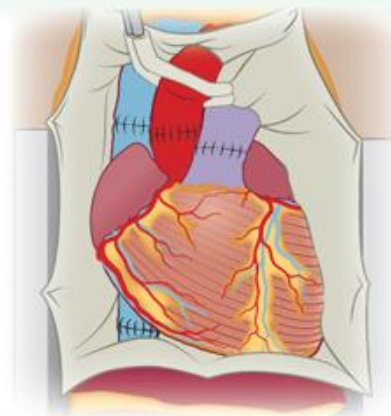


Chest x-ray: severe pulmonary congestion due to left ventricular systolic or diastolic dysfunction.

LVAD as bridge to transplant or permanent therapy



Cardiac transplantation



- Bridge to transplant or destination therapy
- Complications:
 - Stroke
 - Device malfunction/pump thrombosis
 - Major bleeding
 - Major infection
 - Arrhythmia

Chapter 52 Fuster and Hurst's Central Illustration. In many patients with systolic heart failure will ultimately experience worsening dyspnea, exertional intolerance, frequent hospitalizations, and end-organ dysfunction despite receiving guideline-directed medical therapy. As the disease progresses, a left ventricular assist device (LVAD) may be required to support the patient as a bridge to transplant or as permanent therapy. A subset of those with advanced heart failure will be candidates for cardiac transplantation.

Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e
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Health Disparities in HF

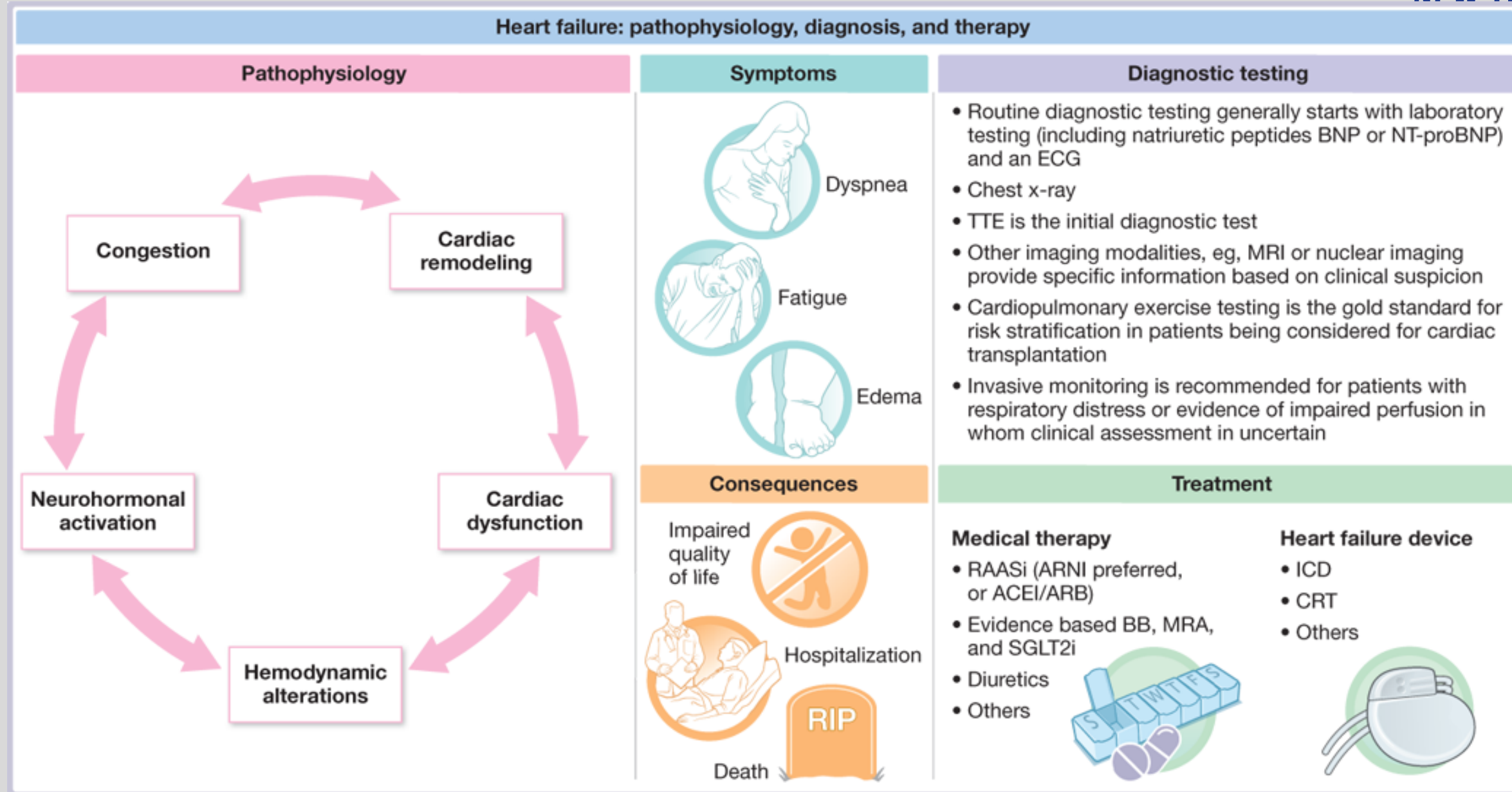
- Highest incidence in self-identified Black patients (HTN)
- HF hospitalization and mortality rates for Black patients are also higher than for White
- 50,000-350,000 immigrants from Mexico and Central America may have *Trypanosoma cruzi*, with 20% progressing to Chagas cardiomyopathy
- Southeast Asian and Pacific Islanders: high prevalence of DM, associated with poor HF outcomes
- Older adults: low income, social isolation, and lack of caregiver support increase HF mortality
- Health care system factors
 - Women are less likely to receive discharge instructions for HF, less likely to be referred to specialty care, and less likely to receive a heart transplantation, compared with men;
 - Black pts are less likely to receive care from a cardiologist during an ICU admission for HF, have less access to specialized inpatient HF care, and may be vulnerable to clinician biases during evaluation for advanced HF therapies;
 - Hispanic patients are disproportionately noninsured, may experience language barriers to quality care, and also have less access to specialized inpatient HF care;
 - Native Americans and Alaskan Natives experience particular challenges in specialty care access because Indian Health Service facilities are often small and rural.

Our Patient Case (revisited)

- A 65-year-old man with a history of HTN, hyperlipidemia, and tobacco use reports to the clinic because of 2 weeks of increasing dyspnea on exertion. He now sleeps on 4 pillows at night to aid in breathing. He has T2DM variably controlled with metformin and diet, and drinks 3-4 glasses of wine/day. Physical examination: jugular venous distention, an S_3 heart sound, and bilateral LE edema. ECG shows Q waves in leads V_2 - V_4 . He gives a history of hospitalization for "issues with my heart" about a year ago when he had sudden chest pain that radiated to his left arm, after which he had been asymptomatic.
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Summary



Chapter 48 Fuster and Hurst's Central Illustration. Heart failure is a clinical syndrome that involves a complex interplay between myocardial, vascular, neurohormonal, hemodynamic and comorbid factors. Signs and symptoms of heart failure can be fairly nonspecific and thus diagnosis of heart failure is often challenging. However, early diagnosis and treatment of heart failure can improve quality of life, and reduce risk of hospitalization and/or death. BB, beta blockers; BNP, B-type natriuretic peptide; CRT, cardiac resynchronization therapy; ECG, electrocardiogram; ICD, implantable cardioverter defibrillator; MRA, mineralocorticoid receptor antagonist; NT-proBNP, N-terminal-pro B-type natriuretic peptide; RAASi, renin-angiotensin-aldosterone system inhibitors; SGLT2i, sodium-glucose co-transporter 2 inhibitors; TTE, transthoracic echocardiography.